Newtonian Cooling and the Adiabatic Index

PY2107 – Experiment 4

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1 Abstract

This experiment investigates Newtonian cooling of a heated calorimeter and determines the adiabatic index of air from the oscillation period of a ball above an enclosed air volume. A linear fit to the logarithm of the cooling temperature excess gives $k = (1.465 \pm 0.007) \times 10^{-4}\,\mathrm{s}^{-1}$, corresponding to an effective air boundary-layer thickness of $b = (4.11 \pm 0.04)\,\mathrm{mm}$. The oscillation measurements give $\gamma = 1.39 \pm 0.08$, consistent within uncertainty with the diatomic ideal-gas prediction of 1.40. The uncertainty in γ is dominated by the uncertainty in the mean oscillation period.

2 Theory

2.1 Newtonian Cooling

When a hot object is surrounded by a cooler fluid, thermal energy is transferred from the object to the surroundings. In this experiment, the calorimeter loses heat to the surrounding air [3, 2].

The laboratory model assumes that heat is conducted through a thin layer of approximately stationary air adjacent to the calorimeter. Outside this boundary layer, convection carries the warmed air away and replaces it with cooler air.

If the boundary layer has thickness b, the heat-transfer rate is approximated by

$$\dot{Q} = \frac{\lambda A}{b}(T - T_0), \tag{1}$$

where λ is the thermal conductivity of air, A is the exposed surface area of the calorimeter, T is its temperature, and T_0 is the ambient temperature.

The model assumes that the temperature difference across the boundary layer is approximately steady and that the calorimeter and water can be represented by a single uniform temperature.

The rate of change of thermal energy of the calorimeter is related to its temperature change by

$$dQ = C dT,$$

where C is the total heat capacity. Since the object loses heat when $T > T_0$, its energy balance is

$$C\frac{dT}{dt} = -\frac{\lambda A}{b}(T - T_0). \tag{2}$$

Defining the temperature excess as

$$\Theta = T - T_0,$$

and assuming that the ambient temperature remains constant gives

$$\frac{d\Theta}{dt} = -k\Theta, \tag{3}$$

where

$$k = \frac{\lambda A}{bC}. \tag{4}$$

Separating variables and integrating,

$$\frac{d\Theta}{\Theta} = -k\,dt,$$

gives

$$\ln\Theta = -kt + \text{constant}.$$

If the initial temperature at $t = 0$ is T_i, then $\Theta_i = T_i - T_0$, and the solution becomes

$$\boxed{T(t) = T_0 + (T_i - T_0)e^{-kt}}. \tag{5}$$

Therefore, the temperature difference between the calorimeter and the surroundings decreases exponentially with time.

The characteristic cooling time is defined as

$$\boxed{\tau = \frac{1}{k}}.$$

A large value of k corresponds to faster cooling, whereas a small value corresponds to slower cooling.

Taking the natural logarithm of Eq. (5) gives

$$\boxed{\ln(T - T_0) = \ln(T_i - T_0) - kt}. \tag{6}$$

Thus, a graph of $\ln(T - T_0)$ against time should be linear, with gradient $-k$. The cooling coefficient can therefore be obtained from the negative gradient of a linear regression.

2.2 Heating and Cooling Simultaneously

During the heating phase, electrical energy is supplied to the calorimeter while heat is simultaneously lost to the surroundings.

The electrical power supplied by the heater is

$$P = VI, \tag{7}$$

where V is the voltage across the heater and I is the current.

The energy balance during heating is

$$C\frac{dT}{dt} = P - \frac{\lambda A}{b}(T - T_0).$$

Using $k = \lambda A/(bC)$ gives

$$\frac{dT}{dt} = \frac{P}{C} - k(T - T_0). \tag{8}$$

For constant power and ambient temperature, with initial calorimeter temperature T_i, the solution is

$$\boxed{T(t) - T_0 = (T_i - T_0)e^{-kt} + \frac{P}{Ck}(1 - e^{-kt})}. \tag{9}$$

This equation predicts that the temperature rises towards a steady-state value when heating and cooling occur simultaneously.

If cooling is negligible over a short interval, the energy balance reduces to

$$C\frac{dT}{dt} = P = VI.$$

Integrating gives

$$\boxed{T(t) = T_i + \frac{VI}{C}t}.$$

Consequently, the temperature should increase approximately linearly during the initial stages of heating if the electrical power is constant and heat losses are small.

In practice, deviations from linearity are expected because heat is lost to the surroundings throughout the heating phase.

2.3 Total Heat Capacity

The total heat capacity is approximated by the sum of the heat capacities of the aluminium calorimeter and the water:

$$\boxed{C = mc_{\text{Al}} + Mc_{\text{water}}}. \tag{10}$$

Here, m is the mass of the aluminium calorimeter, M is the mass of the water, and the specific heat capacities are

$$c_{\text{Al}} = 900\,\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}, \tag{11}$$

$$c_{\text{water}} = 4180\,\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}. \tag{12}$$

All masses must be expressed in kilograms. The resulting total heat capacity has units of $\mathrm{J} \cdot \mathrm{K}^{-1}$.

The stirrer and other components may also absorb heat. Their contributions should be included if their heat capacities are known and significant.

2.4 Air Boundary-Layer Thickness

Rearranging Eq. (4) gives the effective thickness of the air boundary layer:

$$\boxed{b = \frac{\lambda A}{kC}}. \tag{13}$$

The thermal conductivity of dry air is taken to be

$$\lambda = 0.026\,\mathrm{W} \cdot \mathrm{m}^{-1} \cdot \mathrm{K}^{-1}. \tag{14}$$

Once A, k, and C have been measured or calculated, the boundary-layer thickness can be determined.

The result represents an effective thickness within the simplified heat-transfer model rather than a direct geometrical measurement of the air surrounding the calorimeter.

2.5 Adiabatic Index

The adiabatic index of a gas is defined as the ratio of its heat capacity at constant pressure to its heat capacity at constant volume [3]:

$$\boxed{\gamma = \frac{C_P}{C_V}}. \tag{15}$$

For an ideal gas undergoing an adiabatic process, no heat is exchanged with the surroundings. The pressure and volume are related by

$$\boxed{PV^{\gamma} = \text{constant}}. \tag{16}$$

In this experiment, the adiabatic index of air is determined by measuring the period of oscillation of a steel ball or cylinder inside a vertical tube connected to a vessel containing air.

The ball acts as a movable piston. Small vertical displacements of the ball change the volume of the enclosed air, producing pressure changes that provide a restoring force.

2.6 Derivation of the Period of Oscillation

Let m be the mass of the ball, A its cross-sectional area, V the equilibrium volume of the enclosed air, and P the equilibrium pressure of the enclosed air.

At equilibrium, the upward force due to the enclosed air balances the atmospheric pressure acting downward and the weight of the ball. Therefore,

$$PA = P_{\text{atm}}A + mg, \tag{17}$$

which gives

$$\boxed{P = P_{\text{atm}} + \frac{mg}{A}}. \tag{18}$$

Now suppose the ball is displaced upwards by a small distance x. The volume of the enclosed air increases by

$$\Delta V = Ax. \tag{19}$$

The new volume is therefore $V + Ax$. Assuming the oscillations are sufficiently rapid to be adiabatic, Eq. (16) gives

$$P(V + Ax)^{\gamma} = P_0 V^{\gamma}, \tag{20}$$

where P_0 is the equilibrium pressure. Since $P_0 = P$,

$$P(x) = P\left(\frac{V}{V + Ax}\right)^{\gamma}. \tag{21}$$

For small displacements, $Ax \ll V$, the binomial approximation $(1 + u)^{-\gamma} \approx 1 - \gamma u$ gives

$$P(x) \approx P\left(1 - \frac{\gamma Ax}{V}\right). \tag{22}$$

Hence, the pressure change is

$$\Delta P \approx -\frac{\gamma PA}{V}x. \tag{23}$$

The resulting change in the upward force on the ball is $A\Delta P$. The constant forces due to atmospheric pressure and gravity cancel at equilibrium, so Newton's second law gives

$$m\frac{d^2x}{dt^2} = A\Delta P. \tag{24}$$

Substituting the pressure change,

$$m\frac{d^2x}{dt^2} = -\frac{\gamma PA^2}{V}x. \tag{25}$$

Rearranging,

$$\boxed{\frac{d^2x}{dt^2} + \frac{\gamma PA^2}{mV}x = 0}. \tag{26}$$

This is the equation of simple harmonic motion, with angular frequency

$$\omega^2 = \frac{\gamma P A^2}{mV}. \tag{27}$$

Since $\omega = 2\pi/T$, where T is the period of oscillation,

$$\frac{4\pi^2}{T^2} = \frac{\gamma P A^2}{mV}. \tag{28}$$

Rearranging gives the expression used to calculate the adiabatic index:

$$\boxed{\gamma = \frac{4\pi^2 mV}{PA^2T^2}}. \tag{29}$$

All quantities must be expressed in SI units: m in kilograms, V in cubic metres, P in pascals, A in square metres, and T in seconds.

2.7 Expected Value of the Adiabatic Index

Air consists primarily of nitrogen and oxygen, which are diatomic molecules. At ordinary temperatures, a diatomic ideal gas has three translational and two rotational degrees of freedom. Vibrational degrees of freedom are approximately inactive at these temperatures.

The equipartition theorem gives the molar heat capacity at constant volume as

$$C_V = \frac{f}{2}R, \tag{30}$$

where f is the number of active degrees of freedom and R is the molar gas constant.

For a diatomic gas, $f = 5$ [3], so

$$C_V = \frac{5}{2}R. \tag{31}$$

For an ideal gas,

$$C_P - C_V = R. \tag{32}$$

Therefore,

$$C_P = \frac{7}{2}R, \tag{33}$$

and the expected adiabatic index is

$$\boxed{\gamma_{\text{theory}} = \frac{C_P}{C_V} = \frac{7}{5} = 1.40}. \tag{34}$$

This value provides the theoretical reference against which the experimental result can be compared.

3 Experimental Procedure

3.1 Newtonian Cooling

The apparatus consisted of an aluminium calorimeter, water, a stirrer, an electrical heating coil, a power supply, a digital ammeter and voltmeter, two thermometers, a balance, and a computer-based temperature recording system.

The first thermometer recorded the temperature of the calorimeter and water, while the second thermometer measured the ambient-air temperature. Allowing us to dynamically calculate ΔT

The mass m of the empty metal calorimeter was measured using a balance. The curved surface area A of its walls was also determined.

The calorimeter was approximately half-filled with water. The mass M of the water was measured, ensuring that the heating coil was covered.

The lid was fitted, with the thermometer in contact with the water. The apparatus was allowed to settle before the initial water and ambient-air temperatures were recorded.

The heating circuit was connected to the power supply. The ammeter range was checked to ensure that its maximum reading was set to 10 A or less. The electrical power was calculated using $P = VI$.

The water was stirred gently before readings to maintain a reasonably uniform temperature. Care was taken to prevent the stirrer from touching the heating coil.

The LoggerPro software was set up to take measurements every 12 seconds.

After 30 minutes, the heating circuit was disconnected. Temperature measurements continued at 12 second intervals during cooling for an hour and a half.

3.2 Adiabatic Index

The apparatus consisted of a large vessel containing enclosed air, a precision vertical glass tube, and a cylinder that fitted closely inside the tube.

The apparatus was checked to ensure that the tap was closed and that the protective mat was positioned at the bottom of the vessel.

The ball was carefully dropped into the glass tube, allowing it to oscillate vertically on the enclosed air cushion.

The motion was recorded using a camera's slow-motion mode and analysed to determine the oscillation period. The elapsed time measured from the footage was divided by 8 to convert it to real time, accounting for the camera's slow-motion factor.

The time taken for a number of complete oscillations was measured. The period was calculated by dividing the total measured time by the number of oscillations.

Several measurements were taken to obtain repeated estimates of the period. The number of oscillations used for each measurement was chosen so that the period could be determined accurately.

The mass, cross-sectional area, and enclosed air volume were obtained from the ex-

perimental apparatus or measurements. The equilibrium pressure was calculated using Eq. (18). Ambient air pressure was taken from the surface pressure charts for that day, see Fig. 9.

The adiabatic index was calculated from the measured period using Eq. (29).

The ball and tube were handled carefully because the fit between them was precise. Dirt or thermal expansion of the ball could affect its motion and introduce systematic errors.

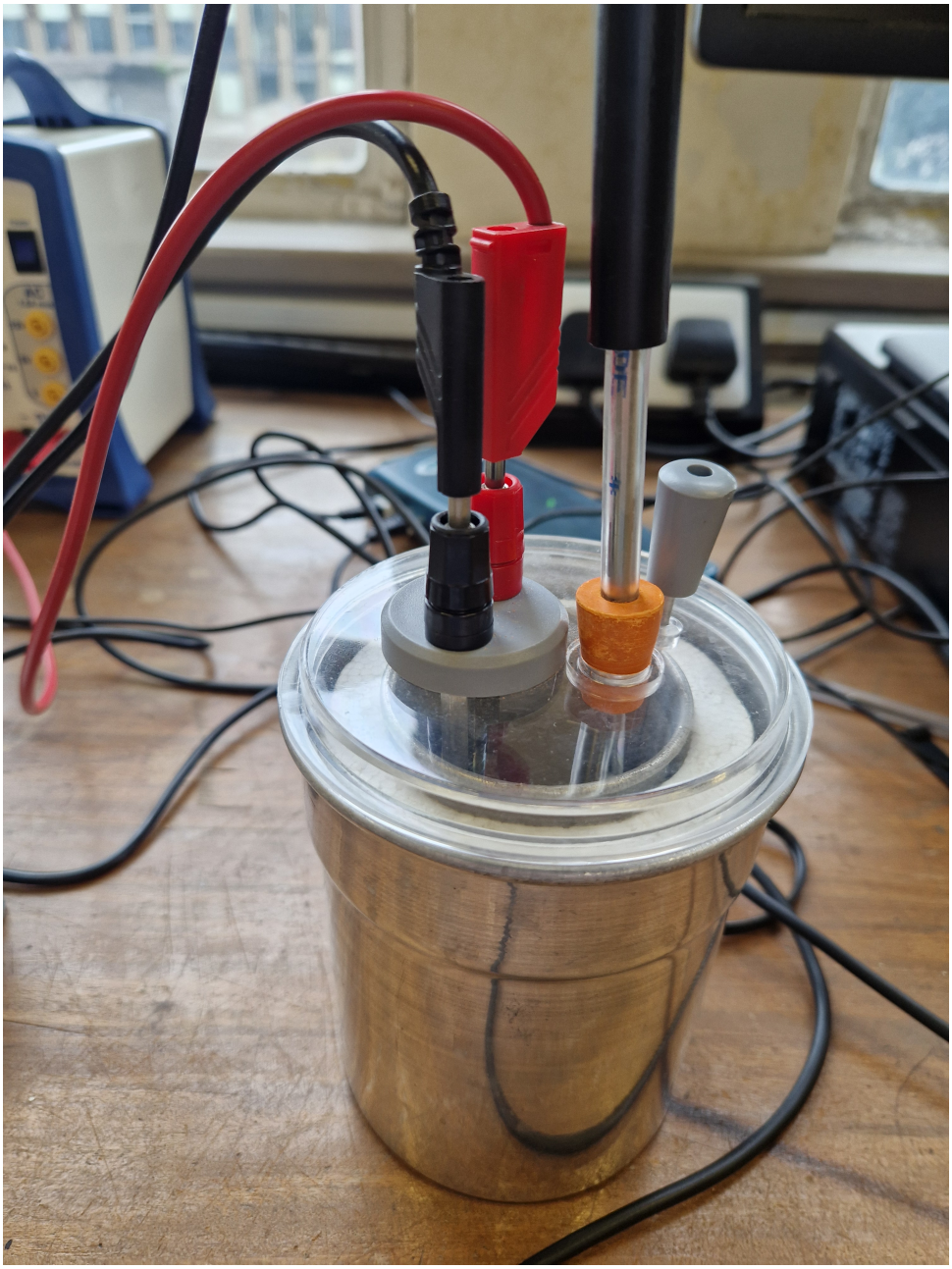


Figure 1: Calorimeter apparatus used in the experiment.

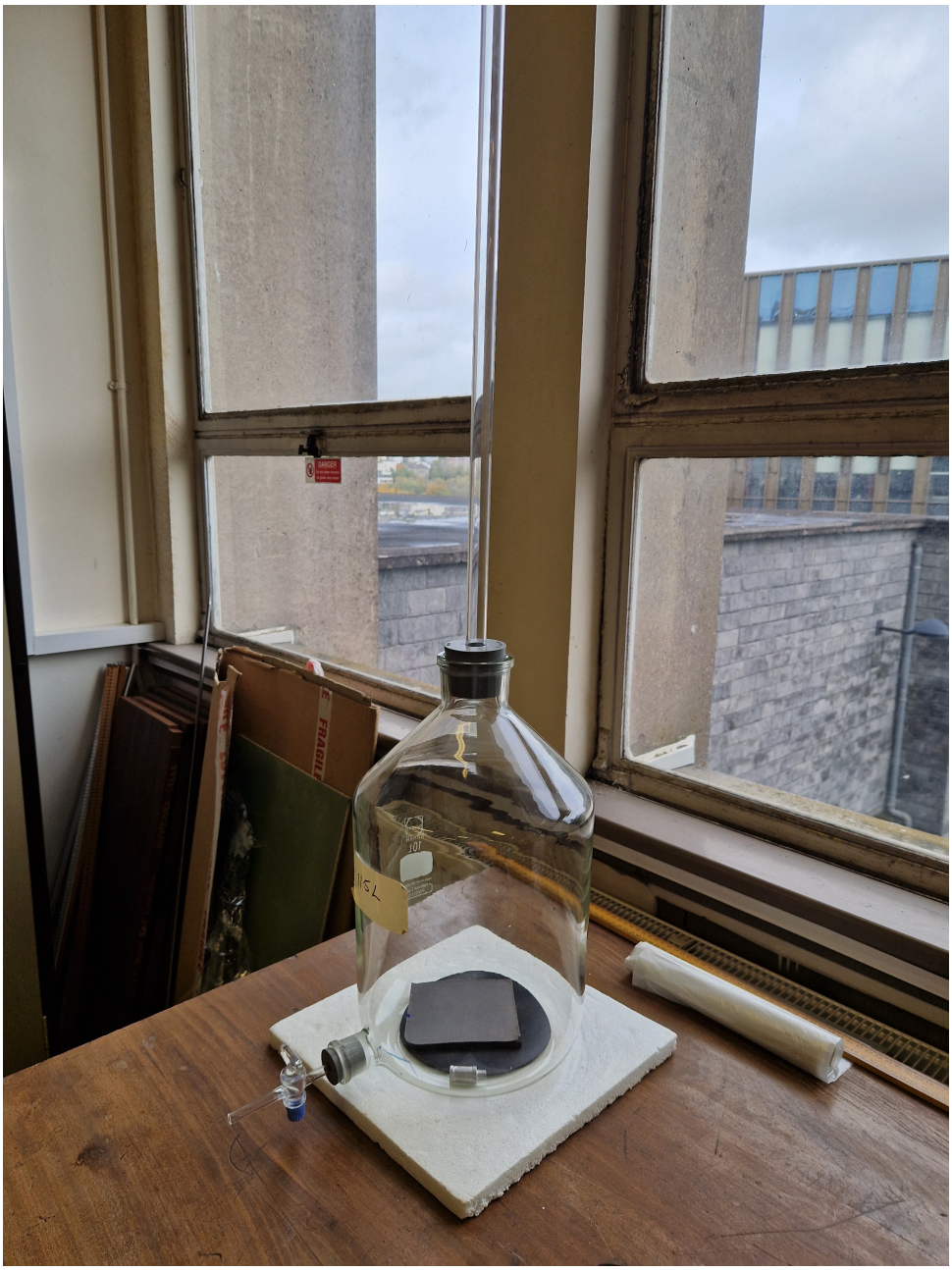


Figure 2: Adiabatic apparatus used in the experiment.

4 Results

4.1 Measured Quantities

The measured quantities used in the analysis are summarised in Table 1.

Table 1: Measured quantities used to calculate k.

Quantity	Symbol	Measured value
Mass of empty calorimeter	m	$0 \cdot 003489\text{pm}0 \cdot 0000005kg$
Mass of water	M	$0 \cdot 150061\text{pm}0 \cdot 000005kg$
Curved surface area	A	$0 \cdot 0145947\text{pm}0 \cdot 000001m^2$
Supply voltage	V	$2 \cdot 98\text{pm}0 \cdot 005V$
Heating current	I	$1 \cdot 25\text{pm}0 \cdot 000005A$

Table 2: Parameters required to calculate γ.

Quantity	Symbol	Value
Mass of ball	m	$0 \cdot 01482\text{pm}0 \cdot 0000005kg$
Cross-sectional area	A	$0 \cdot 000201\text{pm}0 \cdot 000001m^2$
Enclosed air volume	V	$0 \cdot 01\text{pm}0 \cdot 0001m^3$
Atmospheric pressure	P_{atm}	$99600\text{pm}50Pa$
Equilibrium pressure	P	$100323 \cdot 3\text{pm}50Pa$
Mean oscillation period	$\bar{T}$	$1 \cdot 017\text{pm}0 \cdot 03s$

4.2 Temperature-Time Graph

Figure 3 shows the measured temperature difference against elapsed time during the cooling period.

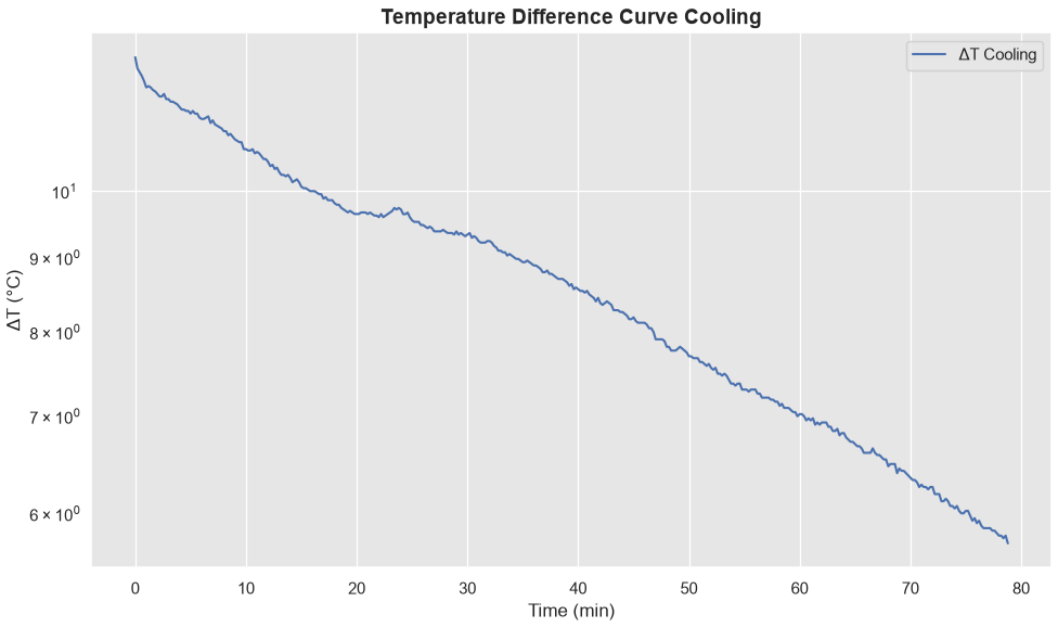


Figure 3: Calorimeter temperature difference as a function of elapsed time for cooling period. Logarithmic scale used for the temperature difference to show linear relationship.

5 Analysis

5.1 Electrical Power

The electrical power supplied to the heater is calculated using

$$P = VI. \tag{35}$$

For example, the calculation using the measured values is

$$P = (2.98)(1.25) = 3.725 \text{ W}.$$

The current was stable as a result of the low voltage and thus temperature.

5.2 Total Heat Capacity

Using Eq. (10),

$$C = mc_{\text{Al}} + Mc_{\text{water}}. \tag{36}$$

Substituting the measured masses,

$$C = (0.0035)(900\,\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}) + (0.15)(4180\,\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}) = 630.4\text{J}\cdot\text{K}^{-1}.$$

5.3 Heating-Rate Comparison

Figure 4 compares the measured temperature excess with a linear fit, a no-loss reference, and the constant-power model including Newtonian cooling. For the latter, $P = VI$ and C are calculated from the measured electrical and calorimeter parameters, while k is taken from the cooling-phase fit; the model starts at the first measured heating excess. With $I = 1.25\,\text{A}$, the no-loss heating rate is $VI/C = 5.909 \times 10^{-3}\,\text{K}\cdot\text{s}^{-1}$, while the measured linear-fit slope is $4.701 \times 10^{-3}\,\text{K}\cdot\text{s}^{-1}$. The lower rate is consistent with heat loss during heating. The cooling-aware curve is the more appropriate ideal prediction for the full heating interval, although its constant-k, constant-power assumptions and the structured residuals limit how closely it should be expected to match every measurement.

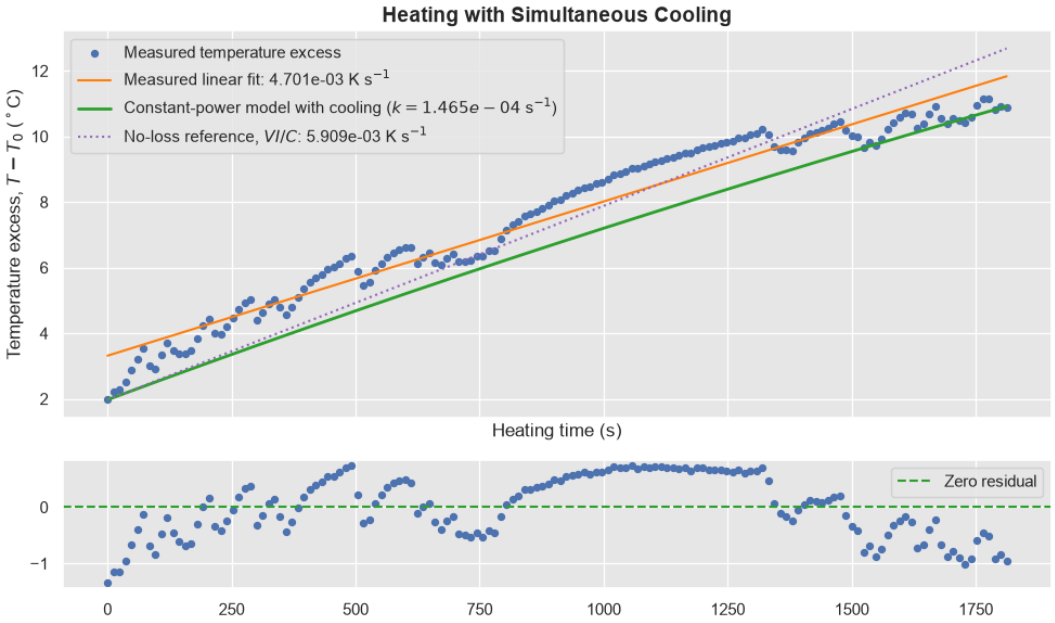


Figure 4: Measured heating temperature excess compared with a linear fit, the constant-power Newton-cooling model using the measured cooling coefficient, and a no-loss reference. The lower panel shows residuals from the cooling-aware model.

5.4 Cooling Coefficient

Let t_c denote the time measured from when the heater was switched off. For every cooling measurement, calculate the temperature excess

$$\Theta = T - T_0. \tag{37}$$

Then calculate $\ln(\Theta)$ for measurements with $\Theta > 0$.

Plotting $\ln(T - T_0)$ against t_c. From Eq. (6) yields a straight line with slope $-k$ as seen in Fig. 5.

The graph yielded

$$k = (1.465 \pm 0.007) \times 10^{-4}\ \text{s}^{-1}, \quad \tau = \frac{1}{k} = (6829 \pm 33)\ \text{s}.$$

A suitable linear fit should be selected by examining the graph and residuals. Measurements very close to ambient temperature may be unreliable because small temperature errors produce large relative errors in $T - T_0$.

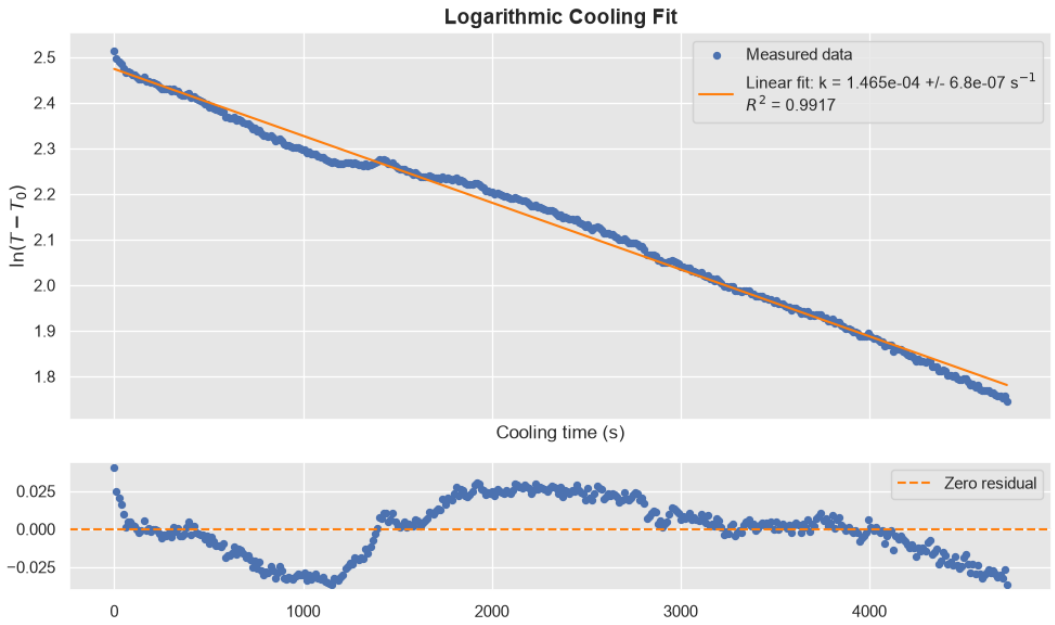


Figure 5: Natural logarithm of the cooling temperature excess against time, with the linear regression used to determine k.



Figure 6: Residuals from the logarithmic cooling fit. A pattern in the residuals indicates departures from a single-exponential cooling model.

The residuals from the full-record fit show systematic structure rather than random scatter. Therefore, the high R^2 value alone does not establish that a single exponential describes the complete cooling interval. The fitted slope uncertainty accounts for regression scatter, but not this model limitation; the fit interval should be checked before treating the resulting k as definitive.

5.5 Approximate Linear Cooling Method

For sufficiently small kt_c, the exponential may be expanded using a Taylor series:

$$e^{-kt_c} \approx 1 - kt_c. \tag{38}$$

Therefore,

$$T_{\text{cool}}(t_c) \approx T_{\max} - (T_{\max} - T_0)kt_c. \tag{39}$$

A graph of temperature against cooling time is consequently approximately linear over a sufficiently short interval.

If its gradient is g, then

$$g \approx -k(T_{\max} - T_0), \tag{40}$$

and hence

$$\boxed{k \approx -\frac{g}{T_{\max} - T_0}}. \tag{41}$$

This is an approximation. The logarithmic fit is the more direct method for estimating k from the exponential cooling model.

5.6 Boundary-Layer Thickness

Using Eq. (13),

$$b = \frac{\lambda A}{kC}. \tag{42}$$

Substituting the thermal conductivity of air and the measured and calculated quantities gives

$$b = (4.11 \pm 0.04)\text{mm}.$$

5.7 Adiabatic Index

Using Eq. (29) with the mean period and apparatus parameters in Table 2,

$$\begin{aligned}\gamma &= \frac{4\pi^2(0.01482)(0.010)}{(100323.3)(0.000201)^2(1.01737)^2} \\ &= 1.395 \approx 1.39.\end{aligned}$$

The uncertainty in this value is propagated from the measured quantities in the Error Analysis section, giving $\gamma = 1.39 \pm 0.08$.

5.8 Fluid Simulation

A supplementary finite-difference fluid simulation was conducted to illustrate buoyancy-driven flow and heat transfer around a cooling calorimeter. It uses a pressure-projection flow solver, explicit temperature advection and diffusion, and a lumped calorimeter energy balance. The simulation is qualitative context; it was not used to calculate the experimental results.

5.8.1 Analytic estimate from geometry

The geometry does permit an analytic estimate. The flow is buoyancy-driven, so the Rayleigh number is the controlling parameter rather than the Reynolds number: $\mathrm{Ra}_H = g\beta\Delta T H^3/(\nu\alpha)$. For air at the film temperature ($\nu = 1.6\times10^{-5}$, $\alpha = 2.25\times10^{-5}\,\mathrm{m^2\cdot s^{-1}}$, $\mathrm{Pr} = 0.71$, $\beta = 1/T_f$) and an initial excess $\Delta T \approx 12.3\,\mathrm{K}$, $\mathrm{Ra}_H \approx 1.3\times10^6$. The Churchill–Chu correlation for a vertical surface [2],

$$\mathrm{Nu}_H = \left\{0.825 + \frac{0.387\,\mathrm{Ra}_H^{1/6}}{\left[1+(0.492/\mathrm{Pr})^{9/16}\right]^{8/27}}\right\}^2 \approx 17.7,$$

gives $h = \mathrm{Nu}_H\lambda/H \approx 4.4\,\mathrm{W\cdot m^{-2}\cdot K^{-1}}$ and an effective conductive layer thickness

$$b_{\mathrm{theory}} = \frac{\lambda}{h} = \frac{H}{\mathrm{Nu}_H} \approx 5.9\,\mathrm{mm}.$$

As the calorimeter cools to $\Delta T \approx 6\,\mathrm{K}$ this rises to about 7 mm. This is directly comparable to the experimental $b = (4.11 \pm 0.04)\,\mathrm{mm}$, since both are defined by $b = \lambda/h$: the analytic value is approximately 44–70% larger. Plausible causes are the neglected radiative loss, the lid and base contributing to heat loss (the experiment's A covers only the curved wall), the assumed H, and the approximate air properties. The Churchill–Chu result also carries roughly ±10–20% correlation uncertainty. The analytic natural-convection estimate provides a separate correlation-based comparison, not a validation of the simulation or a direct prediction of the experimental cooling coefficient.

5.8.2 Simulation diagnostics

For the reported run, the time step was $dt = 0.1$. The code returned $T_{\mathrm{cal}} = 43.55$, a mean fluid temperature of 20.10, and a peak speed of 0.01758, all in the model's grid-based units. The corresponding diagnostics were $\mathrm{Re}_{\mathrm{peak}} = 5.624$, $\mathrm{Pr} = 3.333$, $\mathrm{Gr} = 38592$, $\mathrm{Ra} = 128641$, and a grid-based Nusselt estimate of 18.889. The reported thermal boundary-layer diagnostic was 1.899 grid units, and the fitted cooling time constant was 11.82 simulation-time units.

These are diagnostics of the numerical run, not measurements of the laboratory apparatus. In particular, the grid-based thermal boundary-layer diagnostic is not the effective conduction thickness $b = \lambda/h$ inferred from the experiment, and the simulation has no dimensional calibration to metres or seconds.

5.8.3 Convective heat loss and qualitative analysis

Figs. 7 and 8 show the convective heat loss and Newton cooling curves for the calorimeter. The simulation illustrates heat transfer accompanied by buoyancy-driven flow. It does not establish that the simulation quantitatively reproduces the laboratory cooling curve.

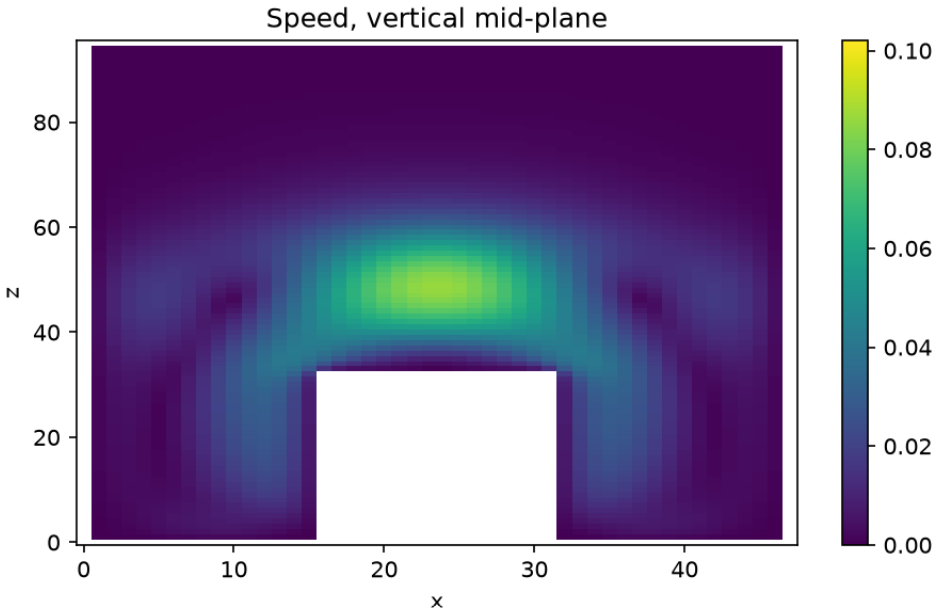


Figure 7: Convective heat loss from the calorimeter.

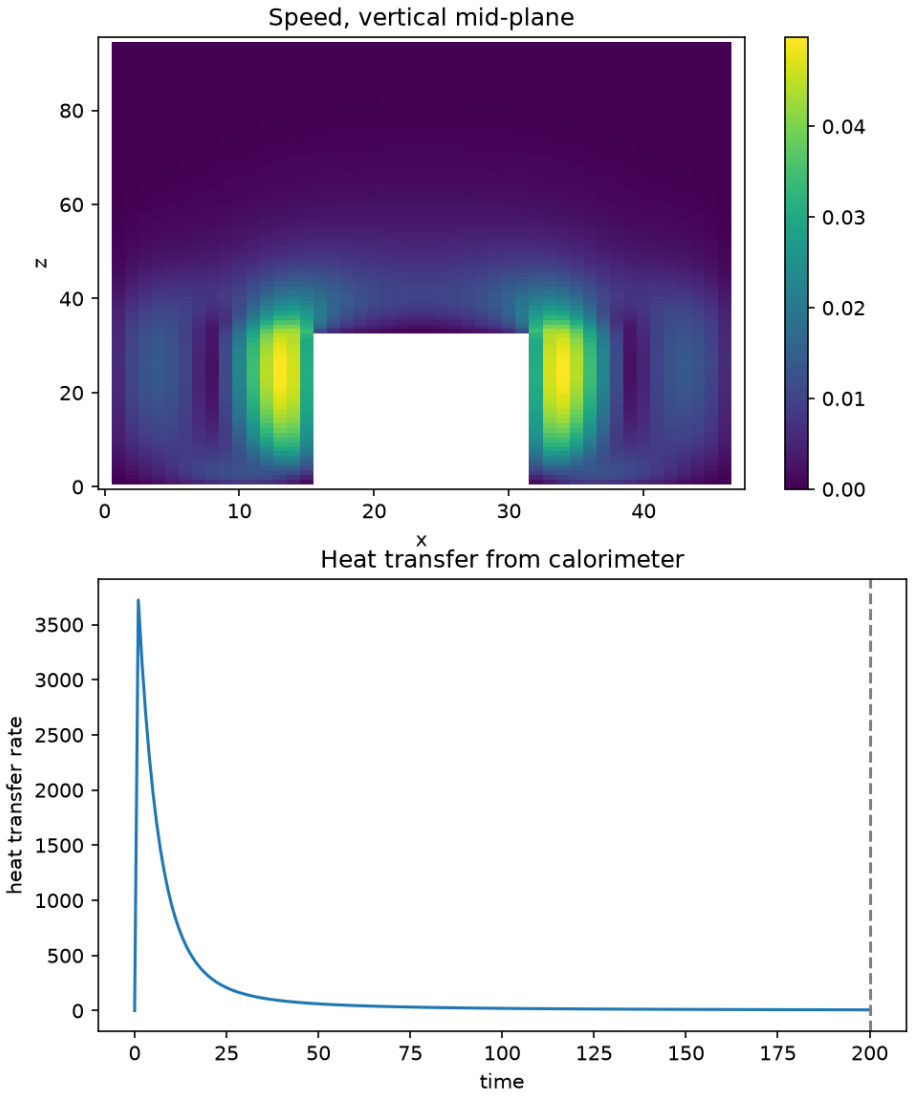


Figure 8: Simulation of convective heat loss from the calorimeter.

6 Error Analysis

The principal sources of uncertainty in the cooling analysis are the mass measurements, the calorimeter surface area, the temperature readings, the ambient temperature, and the fitted cooling coefficient.

The uncertainty in k directly affects the inferred boundary-layer thickness.

For independent uncertainties,

$$\left(\frac{\sigma_b}{b}\right)^2 = \left(\frac{\sigma_A}{A}\right)^2 + \left(\frac{\sigma_k}{k}\right)^2 + \left(\frac{\sigma_C}{C}\right)^2, \tag{43}$$

where uncertainty in λ has been neglected.

The uncertainty in heat capacity is

$$\sigma_C = \sqrt{(c_{\mathrm{Al}}\sigma_m)^2 + (c_{\mathrm{water}}\sigma_M)^2}. \tag{44}$$

The absolute uncertainty in the boundary-layer thickness is then

$$\boxed{\sigma_b = b\sqrt{\left(\frac{\sigma_A}{A}\right)^2 + \left(\frac{\sigma_k}{k}\right)^2 + \left(\frac{\sigma_C}{C}\right)^2}}. \tag{45}$$

$$\boxed{b = (4.11 \pm 0.04)\mathrm{mm}}.$$

Other sources of systematic error include heat transfer through the lid and supports, imperfect mixing of the water, sensor response time, and the assumption that the boundary layer has a constant thickness.

6.1 Adiabatic Index

Assuming independent uncertainties in the quantities in Eq. (29), propagation of uncertainty gives

$$\left(\frac{\sigma_\gamma}{\gamma}\right)^2 = \left(\frac{\sigma_m}{m}\right)^2 + \left(\frac{\sigma_V}{V}\right)^2 + \left(\frac{\sigma_P}{P}\right)^2 + \left(2\frac{\sigma_A}{A}\right)^2 + \left(2\frac{\sigma_{\bar{T}}}{\bar{T}}\right)^2. \tag{46}$$

Using the uncertainties in Table 2 gives $\sigma_\gamma \approx 0.084$, so the result is reported as

$$\boxed{\gamma = 1.39 \pm 0.08}.$$

Here, the uncertainty in the mean period is the standard error of the six measured periods. It is the dominant contribution to the propagated uncertainty; the result does not include additional systematic uncertainty from the camera's slow-motion calibration or departures from the ideal adiabatic model.

7 Discussion

The measurements test two separate models: exponential cooling of a calorimeter and approximately adiabatic compression of air by an oscillating piston. A small numerical natural-convection model was also explored as a side project. It helps illustrate a possible flow mechanism during cooling, but it was not part of the laboratory requirements and is not used to calculate the experimental results.

7.1 Heating Phase

For constant electrical power and negligible heat loss, the expected initial heating rate is

$$\frac{dT}{dt} = \frac{VI}{C}. \tag{47}$$

The measured fit gives $4.701 \times 10^{-3}\,\mathrm{K \cdot s^{-1}}$, compared with the no-loss prediction $5.909 \times 10^{-3}\,\mathrm{K \cdot s^{-1}}$. The measured rate is about 20% lower. This is consistent with heat being lost while the calorimeter is heated; structured residuals also show that a single straight line is only an approximate description of the full heating record. The current is recorded as 1.25 A, consistent with the power calculation. Its stated uncertainty is retained from the measurement table, although its very small size should be checked against the instrument specification.

7.2 Cooling Phase

Newtonian cooling predicts an exponential decrease in the temperature difference between the calorimeter and its surroundings [3, 2]. The fit gives $k = (1.465 \pm 0.007) \times 10^{-4}\,\mathrm{s^{-1}}$, corresponding to $\tau = 6829 \pm 33\,\mathrm{s}$. This uncertainty describes the regression fit, not every source of experimental uncertainty.

Although the logarithmic graph is broadly linear, its residuals show systematic structure. Thus, the fit supports an exponential-like trend over the measured interval, but does not establish that one cooling constant describes the entire record. Changes in ambient temperature, multiple heat-transfer paths, and non-uniform calorimeter temperature are plausible contributors. The reported uncertainty should be understood as a statistical fit uncertainty, with additional model uncertainty not quantified here.

7.3 Boundary-Layer Thickness

The resulting $b = (4.11 \pm 0.04)\,\mathrm{mm}$ follows from the single-layer conduction approximation and the fitted k. It is an effective model parameter, not a direct measurement of the fluid's local thermal boundary-layer profile. Its small quoted uncertainty does not include uncertainty in the assumptions of uniform temperature, constant ambient conditions, or the chosen heat-transfer model.

7.4 Adiabatic Index

The measured value, $\gamma = 1.39 \pm 0.08$, is consistent within uncertainty with the diatomic ideal-gas value of 1.40 from equipartition theory [3]. The difference is only 0.006, much smaller than the propagated uncertainty. The uncertainty is dominated by the spread in the measured periods. The camera's slow-motion calibration, enclosed-volume and tube-area measurements, and possible friction or leakage around the ball may add systematic effects not included in that propagation.

7.5 Supplementary Natural-Convection Simulation

The accompanying simulation is a separate numerical demonstration of cooling from a hot, vertical cylindrical calorimeter. It uses a pressure-projection flow solver, explicit temperature advection and diffusion, and a lumped-temperature calorimeter energy balance [1]. The closed outer walls are no-slip and thermally insulated; heat passes between the calorimeter and fluid. The model illustrates buoyancy-driven flow during cooling, but is not evidence that the simulation quantitatively reproduces the laboratory cooling curve. Its run diagnostics are summarised in Appendix 4.

7.6 Limitations

The main limitations of the experiment include:

- Changes in ambient temperature, sensor uncertainty, imperfect mixing, and heat loss through the calorimeter lid and supports.
- Structured residuals in the heating and cooling fits, which indicate that the simple models do not describe every measurement equally well.
- The uniform-temperature and constant-effective-layer assumptions in the calorimeter analysis.
- Timing and slow-motion calibration uncertainty, as well as possible friction or leakage in the oscillation apparatus.
- For the supplementary simulation, no dimensional calibration, grid/time-step convergence tests, or validation against a benchmark or the measured cooling data. The explicit thermal scheme and idealised geometry also limit quantitative conclusions.

The experimental uncertainty estimates account for selected measurement scatter, but not all systematic or model errors. Likewise, the simulation is best treated as qualitative context until its convergence and physical calibration have been established.

8 Conclusion

This experiment investigated Newtonian cooling of a heated calorimeter and determined the adiabatic index of air from the oscillation of a ball above an enclosed air volume.

The cooling fit yielded $k = (1.465 \pm 0.007) \times 10^{-4}\,\mathrm{s}^{-1}$ and an effective air-layer thickness of $b = (4.11 \pm 0.04)\,\mathrm{mm}$. The oscillation analysis gave $\gamma = 1.39 \pm 0.08$, consistent with the diatomic ideal-gas prediction of 1.40. The uncertainty in γ is dominated by the measured period spread, while systematic timing and apparatus effects remain possible.

The measured heating rate was approximately 20% below the constant-power, no-loss prediction, as expected when heat loss occurs during heating. Systematic residual structure in both the heating and cooling analyses cautions against interpreting the quoted fit uncertainties as complete uncertainty budgets.

As a side project, the fluid simulation provided a qualitative example of buoyancy-driven flow coupled to calorimeter cooling: the reported run produced non-zero flow while the calorimeter cooled to a model temperature of 43.55. Its grid-based diagnostics cannot be compared directly with the laboratory measurements, so it supplements rather than changes the experimental conclusions. A dimensional mapping, convergence study, and validation would be needed before making quantitative claims from that model.

9 References

References

[1] Ian Cullinane. *3D Natural Convection Cooling Simulation*. Python source code, unpublished. 2026.

[2] Frank P. Incropera et al. *Fundamentals of Heat and Mass Transfer*. 8th ed. John Wiley & Sons, 2018.

[3] *Physics 2107: Newtonian Cooling and the Adiabatic Index of Air, Experiment 4*. Laboratory handout supplied for PY2107. Department of Physics. 2026.

A Appendix

A.1 Results

For each trial, if N complete oscillations take a measured slow-motion video time t_{video}, the real-time period is

$$\boxed{T = \frac{t_{\mathrm{video}}}{8N}}. \tag{48}$$

The factor of 8 converts the video time to real time. The measured video times, oscillation counts, and calculated periods are recorded in Table 3.

Table 3: Measured oscillation periods.

Trial	Video time, t_{video} (s)	Number of oscillations, N	Period, T (s)
1	26.04	3	1.085
2	52.45	7	0.936607
3	60.45	8	0.944531
4	77.62	10	0.97025
5	34.73	4	1.08531
6	25.98	3	1.0825

A.2 Weather conditions

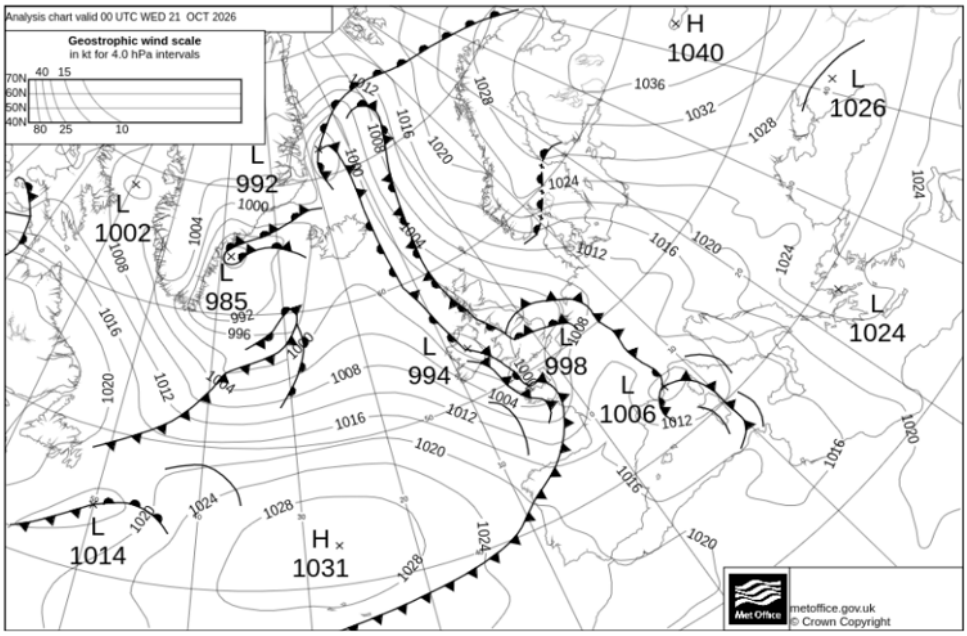


Figure 9: Surface pressure chart showing the weather conditions during the experiment.

A.3 Supplementary simulation: model and run

This simulation was an optional side project, not a requirement of the laboratory briefing. The implementation uses a finite-difference pressure-projection method for flow and advances the thermal field using explicit upwind advection and face-based diffusion. The calorimeter has a spatially uniform temperature and loses heat exchanged at its fluid-facing cells. The outer walls are stationary and impermeable, and thermally insulated. This is a simplified natural-convection model; the code is cited as the implementation source [1].

The reported run used $dt = 0.1$. Its other configuration parameters are not specified here, so the diagnostics below are reported as returned by the simulation rather than as a reproducible benchmark. Values are in grid-based or nondimensional units unless otherwise noted.

Table 4: Diagnostics returned by the reported supplementary simulation run.

Diagnostic	Value
Time step, dt	0.1
Calorimeter temperature	43.5549
Mean fluid temperature	20.1032
Maximum speed	1.7576×10^{-2}
Peak Reynolds number, Re	5.624
Prandtl number, Pr	3.333
Grashof number, Gr	38592.3
Rayleigh number, Ra	128640.9
Grid-based Nusselt estimate, Nu	18.889
Thermal boundary-layer diagnostic	1.899 grid units
Cooling time constant	11.815 simulation-time units
Relative thermal energy	34678.32

The thermal boundary-layer value is a grid-based diagnostic, not a measurement of the experimental boundary layer or of the effective thickness $b = \lambda/h$. The relative thermal energy is an internal numerical diagnostic, not a substitute for convergence or validation testing.